HDOC DAY-65

1. Aim

To write a recursive C program to solve the Tower of Hanoi problem for n disks and display all the moves required to transfer the disks from the source rod to the destination rod.

2. Problem Statement

Write a recursive program to solve the Tower of Hanoi problem for n disks and display all moves.

Rules:

1. Only one disk can be moved at a time.
2. Only the top disk can be moved from a rod.
3. A larger disk cannot be placed on a smaller disk.

3. Algorithm

1. Start.
2. Read the number of disks n.
3. Define a recursive function towerOfHanoi(n, source, auxiliary, destination).
4. If n == 1, move the disk from source to destination.
5. Otherwise:
    - Move n-1 disks from source to auxiliary.
    - Move the largest disk from source to destination.
    - Move n-1 disks from auxiliary to destination.
6. Display each move.
7. Stop.

4. C Program

```
#include <stdio.h>


void towerOfHanoi(int n, char source,
                  char auxiliary, char destination)
{
   if (n == 1)
   {
      printf("Move disk 1 from %c to %c\n",
```

```
            source, destination);
        return;
    }

    towerOfHanoi(n - 1, source,
                destination, auxiliary);

    printf("Move disk %d from %c to %c\n",
           n, source, destination);

    towerOfHanoi(n - 1, auxiliary,
                source, destination);
}
```